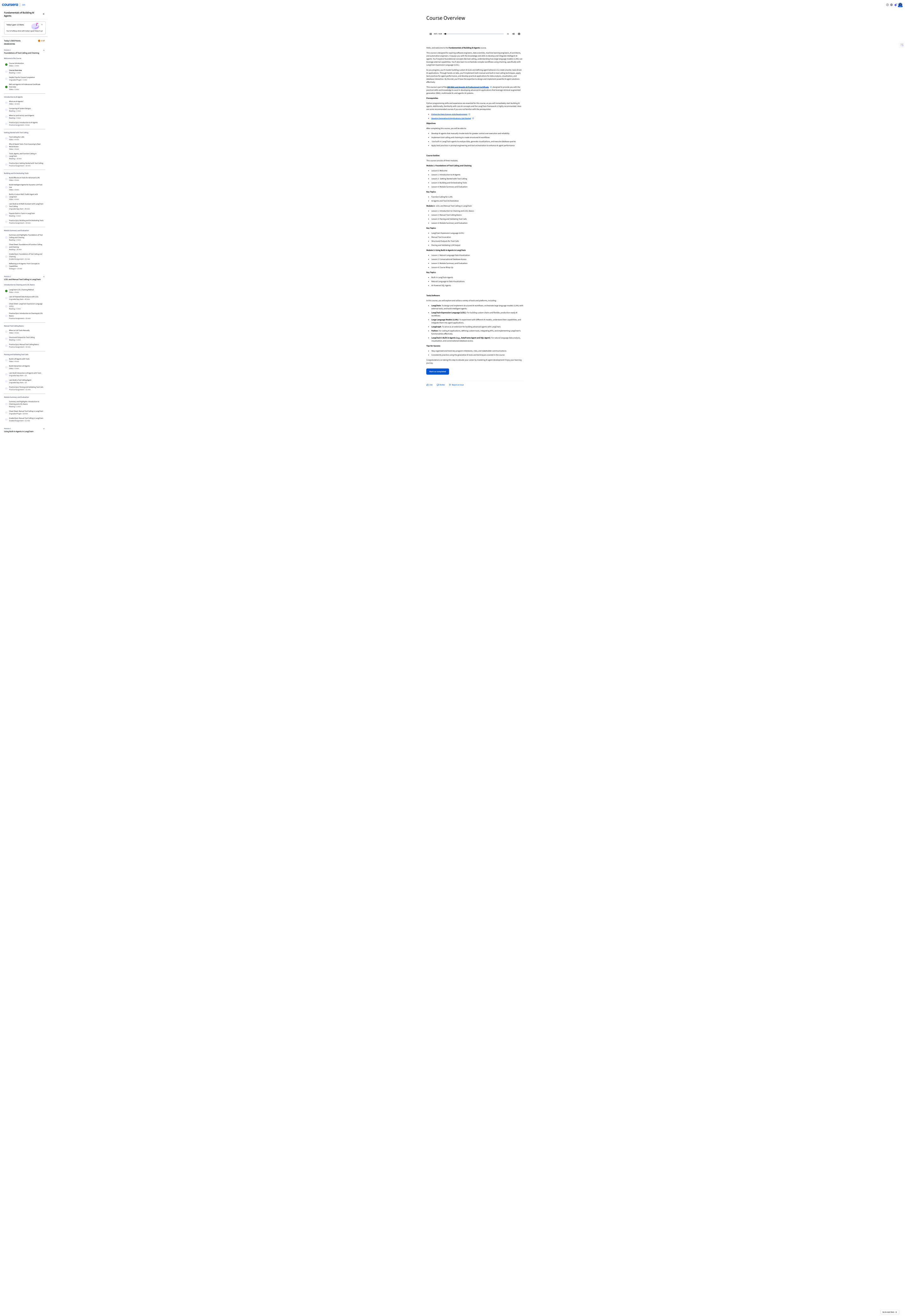

Fundamentals of Building AI Agents

Today's goal • 1/3 items

You're halfway done with today's goal! Keep it up!

Today's Skill Points 0 XP

See skill progress

Module 1

Foundations of Tool Calling and Chaining

Welcome to the Course

- ✓ Course Introduction
  Video • 2 min
- ○ **Course Overview**
  Reading • 3 min
- ○ Helpful Tips for Course Completion
  Ungraded Plugin • 1 min
- ✓ RAG and Agentic AI Professional Certificate Overview
  Video • 5 min

Introduction to AI Agents

- ○ What are AI Agents?
  Video • 12 min
- ○ Comparing AI System Designs
  Reading • 5 min
- ○ When to (and not to) use AI Agents
  Reading • 5 min
- ○ Practice Quiz: Introduction to AI Agents
  Practice Assignment • 9 min

Getting Started with Tool Calling

- ○ Tool Calling for LLMs
  Video • 4 min
- ○ Why AI Needs Tools: From Guessing to Real-World Action
  Video • 4 min
- ○ Tools, Agents, and Function Calling in LangChain
  Reading • 10 min
- ○ Practice Quiz: Getting Started with Tool Calling
  Practice Assignment • 10 min

Building and Orchestrating Tools

- ○ Build Effective AI Tools for Advanced LLMs
  Video • 8 min
- ○ Build Intelligent Agents for Dynamic LLM Tool Use
  Video • 8 min
- ○ Build a Custom Math Toolkit Agent with LangChain
  Video • 6 min
- ○ Lab: Build an AI Math Assistant with LangChain Tool Calling
  Ungraded App Item • 45 min
- ○ Popular Built-in Tools in LangChain
  Reading • 5 min
- ○ Practice Quiz: Building and Orchestrating Tools
  Practice Assignment • 10 min

Module Summary and Evaluation

- ○ Summary and Highlights: Foundations of Tool Calling and Chaining
  Reading • 2 min
- ○ Cheat Sheet: Foundations of Function Calling and Chaining
  Reading • 20 min
- ○ Graded Quiz: Foundations of Tool Calling and Chaining
  Graded Assignment • 21 min
- ○ Reflecting on AI Agents: From Concepts to Capabilities
  Dialogue • 15 min

Module 2

LCEL and Manual Tool Calling in LangChain

Introduction to Chaining and LCEL Basics

- ✓ LangChain LCEL Chaining Method
  Video • 4 min
- ○ Lab: AI Powered Data Analysis with LCEL
  Ungraded App Item • 45 min
- ○ Cheat Sheet: LangChain Expression Langauge (LCEL)
  Reading • 5 min
- ○ Practice Quiz: Introduction to Chaining & LCEL Basics
  Practice Assignment • 10 min

Manual Tool Calling Basics

- ○ When to Call Tools Manually
  Video • 4 min
- ○ Structured Outputs for Tool Calling
  Reading • 5 min
- ○ Practice Quiz: Manual Tool Calling Basics
  Practice Assignment • 10 min

Parsing and Validating Tool Calls

- ○ Build LLM Agents with Tools
  Video • 4 min
- ○ Build Interactive LLM Agents
  Video • 5 min
- ○ Lab: Build Interactive LLM Agents with Tools
  Ungraded App Item • 1h
- ○ Lab: Build a Tool Calling Agent
  Ungraded App Item • 1h
- ○ Practice Quiz: Parsing and Validating Tool Calls
  Practice Assignment • 15 min

Module Summary and Evaluation

- ○ Summary and Highlights: Introduction to Chaining and LCEL Basics
  Reading • 2 min
- ○ Cheat Sheet: Manual Tool Calling in LangChain
  Ungraded Plugin • 20 min
- ○ Graded Quiz: Manual Tool Calling in LangChain
  Graded Assignment • 21 min

Module 3

Using Built-in Agents in LangChain

Course Overview

Hello, and welcome to the **Fundamentals of Building AI Agents** course.

This course is designed for aspiring software engineers, data scientists, machine learning engineers, AI architects, and automation engineers. It equips you with the knowledge and skills to develop and integrate intelligent AI agents. You'll explore foundational concepts like tool calling, understanding how large language models (LLMs) can leverage external capabilities. You'll also learn to orchestrate complex workflows using chaining, specifically with LangChain Expression Language (LCEL).

As you progress, you'll master building custom AI tools and defining agent behaviors to create smarter, task-driven AI applications. Through hands-on labs, you'll implement both manual and built-in tool calling techniques, apply best practices for agent performance, and develop practical applications for data analysis, visualization, and database interaction. By the end, you'll have the expertise to design and implement powerful AI agent solutions effectively.

This course is part of the **IBM RAG and Agentic AI Professional Certificate**, designed to provide you with the practical skills and knowledge to excel in developing advanced AI applications that leverage retrieval-augmented generation (RAG), multimodal AI, and agentic AI systems.

Prerequisites

Python programming skills and experience are essential for this course, as you will immediately start building AI agents. Additionally, familiarity with core AI concepts and the LangChain framework is highly recommended. Here are some recommended courses if you are not familiar with the prerequisites:

- Python for Data Science, AI & Development
- Develop Generative AI Applications: Get Started

Objectives

After completing this course, you will be able to:

- Develop AI agents that manually invoke tools for greater control over execution and reliability
- Implement tool calling and chaining to create structured AI workflows
- Use built-in LangChain agents to analyze data, generate visualizations, and execute database queries
- Apply best practices in prompt engineering and tool orchestration to enhance AI agent performance

Course Outline

This course consists of three modules.

Module 1: Foundations of Tool Calling and Chaining

- Lesson 0: Welcome
- Lesson 1: Introduction to AI Agents
- Lesson 2: Getting Started with Tool Calling
- Lesson 3: Building and Orchestrating Tools
- Lesson 4: Module Summary and Evaluation

Key Topics:

- Function Calling for LLMs
- AI Agents and Tool Orchestration

Module 2: LCEL and Manual Tool Calling in LangChain

- Lesson 1: Introduction to Chaining and LCEL Basics
- Lesson 2: Manual Tool Calling Basics
- Lesson 3: Parsing and Validating Tool Calls
- Lesson 4: Module Summary and Evaluation

Key Topics:

- LangChain Expression Language (LCEL)
- Manual Tool Invocation
- Structured Outputs for Tool Calls
- Parsing and Validating LLM Output

Module 3: Using Built-in Agents in LangChain

- Lesson 1: Natural Language Data Visualization
- Lesson 2: Conversational Database Access
- Lesson 3: Module Summary and Evaluation
- Lesson 4: Course Wrap-Up

Key Topics:

- Built-in LangChain Agents
- Natural Language to Data Visualizations
- AI-Powered SQL Agents

Tools/Software

In this course, you will explore and utilize a variety of tools and platforms, including:

- **LangChain**: To design and implement structured AI workflows, orchestrate large language models (LLMs) with external tools, and build intelligent agents.
- **LangChain Expression Language (LCEL)**: For building custom chains and flexible, production-ready AI workflows.
- **Large Language Models (LLMs)**: To experiment with different AI models, understand their capabilities, and integrate them into agent applications.
- **LangGraph**: To serve as an extension for building advanced agents with LangChain.
- **Python**: For coding AI applications, defining custom tools, integrating APIs, and implementing LangChain's functionalities effectively.
- **LangChain's Built-in Agents (e.g., DataFrame Agent and SQL Agent)**: For natural language data analysis, visualization, and conversational database access.

Tips for Success

- Stay organized and track key program milestones, risks, and stakeholder communications
- Consistently practice using the generative AI tools and techniques covered in the course

Congratulations on taking this step to elevate your career by mastering AI agent development! Enjoy your learning journey.